

1. TWO POINTER

- FIXED DISTANCE POINTERS
 - VARIABLE DISTANCE POINTERS
 - OPPOSITE ENDS (LEFT-RIGHT)
 - SAME DIRECTION POINTERS
 - FAST-SLOW POINTERS (FLOYD'S CYCLE DETECTION)
 - PATTERNS
 - SORTED ARRAY SUM PROBLEMS
 - REMOVE DUPLICATES
 - MERGING SORTED ARRAYS
 - PALINDROME CHECK
-

2. SLIDING WINDOW

- FIXED WINDOW SIZE
- VARIABLE WINDOW SIZE
- MAXIMUM/MINIMUM WINDOW VALUE
- LONGEST SUBSTRING/SUBARRAY PATTERNS
- ANAGRAMS IN STRING

- **PATTERNS**

- MAXIMUM SUM SUBARRAY
 - SMALLEST SUBARRAY $\geq K$
 - LONGEST SUBSTRING WITH K DISTINCT CHARACTERS
-

3. MATRIX

- BASIC TRAVERSALS
 - SPIRAL TRAVERSAL
 - DIAGONAL TRAVERSAL
 - ROTATE / TRANSPOSE
 - SEARCH IN MATRIX (BINARY SEARCH VARIANTS)
 - ISLANDS / FLOOD FILL
 - DP ON MATRIX (MIN PATH, UNIQUE PATHS)
-

4. HASHMAP

- HASHING BASICS
- HASHSET
- FREQUENCY COUNTING
- PREFIX SUM + HASHMAP

- **PATTERNS**

- SUBARRAY SUM = K
 - LONGEST CONSECUTIVE SEQUENCE
 - GROUP ANAGRAMS
-

5. INTERVALS

- MERGING INTERVALS
 - INSERT INTERVAL
 - MINIMUM MEETING ROOMS
 - SWEEP LINE ALGORITHM
 - PATTERNS
 - INTERVAL SCHEDULING
 - OVERLAPPING INTERVALS
 - MINIMUM NUMBER OF ARROWS
-

6. STACK

- BASIC OPERATIONS (PUSH, POP, PEEK)
- NEXT GREATER ELEMENT
- VALID PARENTHESIS

- . INFIX → POSTFIX / PREFIX
 - . EVALUATE POSTFIX/PREFIX
 - . MONOTONIC STACK
 - . HISTOGRAM MAX AREA
-

7. LINKED LIST

- . SINGLY LINKED LIST
 - . DOUBLY LINKED LIST
 - . CIRCULAR LINKED LIST
 - . REVERSAL (ITERATIVE, RECURSIVE)
 - . MERGE TWO SORTED LISTS
 - . DETECT CYCLE
 - . FLATTEN LIST
 - . LRU CACHE
-

8. BINARY TREE (GENERAL)

- . TREE TRAVERSALS (INORDER, PREORDER, POSTORDER)
- . HEIGHT, DIAMETER, DEPTH
- . LOWEST COMMON ANCESTOR

- **SERIALIZATION / DESERIALIZATION**
 - **RECURSION PATTERNS**
-

9. BINARY TREE BFS

- **LEVEL ORDER TRAVERSAL**
 - **ZIG-ZAG TRAVERSAL**
 - **RIGHT/LEFT VIEW**
 - **VERTICAL ORDER TRAVERSAL**
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10. BINARY SEARCH TREE

- **INSERTION, DELETION, SEARCH**
 - **VALIDATE BST**
 - **KTH SMALLEST / LARGEST**
 - **LCA IN BST**
 - **CONVERT SORTED ARRAY TO BST**
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11. GRAPH GENERAL

- **REPRESENTATION (ADJ LIST, MATRIX)**
- **DIRECTED / UNDIRECTED**

- . WEIGHTED / UNWEIGHTED
 - . CONNECTED COMPONENTS
-

12. GRAPH BFS

- . SHORTEST PATH (UNWEIGHTED)
 - . BIPARTITE CHECK
 - . FLOOD FILL
-

13. TRIE

- . BASIC INSERTION, SEARCH, DELETE
 - . PREFIX SEARCH
 - . AUTOCOMPLETE SYSTEM
 - . WORD DICTIONARY
 - . MAXIMUM XOR WITH TRIE
-

14. BACKTRACKING

- . PERMUTATIONS
- . COMBINATIONS
- . SUBSETS

- . N-QUEENS
 - . SUDOKU SOLVER
 - . WORD SEARCH
-

15. MONOTONIC STACK AND QUEUE

- . NEXT GREATER ELEMENT
 - . NEXT SMALLER ELEMENT
 - . SLIDING WINDOW MAXIMUM
 - . STOCK SPAN PROBLEM
-

16. GREEDY

- . INTERVAL SCHEDULING
 - . ACTIVITY SELECTION
 - . JOB SEQUENCING
 - . HUFFMAN ENCODING
 - . MINIMUM SPANNING TREE (KRUSKAL)
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17. DIVIDE AND CONQUER

- . MERGE SORT

- . QUICK SORT
 - . BINARY SEARCH
 - . MAXIMUM SUBARRAY (DIVIDE AND CONQUER)
 - . MATRIX MULTIPLICATION (STRASSEN)
-

18. KADANE'S ALGORITHM

- . MAXIMUM SUBARRAY SUM
 - . 2D KADANE (MATRIX)
-

19. BINARY SEARCH

- . BASIC SEARCH
 - . FIRST/LAST OCCURRENCE
 - . SEARCH IN ROTATED SORTED ARRAY
 - . MEDIAN OF TWO SORTED ARRAYS
 - . BINARY SEARCH ON ANSWER
-

20. MIN AND MAX HEAP

- . HEAPIFY
- . INSERTION/DELETION

- . HEAPSORT
 - . PRIORITY QUEUE IMPLEMENTATION
 - . KTH LARGEST/SMALLEST
 - . MEDIAN IN STREAM
-

21. SEGMENT TREE

- . RANGE SUM QUERY
 - . RANGE MIN/MAX QUERY
 - . LAZY PROPAGATION
 - . UPDATE QUERIES
-

22. FENWICK TREE (BINARY INDEXED TREE)

- . PREFIX SUM QUERY
 - . UPDATE QUERIES
 - . RANGE QUERIES
-

23. B TREE / B+ TREE

- . INSERTION / DELETION
- . SEARCH OPERATION

- . RANGE QUERY (B+ TREE)
-

24. BFS / DFS

- . RECURSIVE DFS
 - . ITERATIVE DFS
 - . BFS TRAVERSAL
 - . SHORTEST PATH
-

25. RADIX

- . RADIX SORT
 - . COUNTING SORT
 - . BUCKET SORT
-

26. BIT MANIPULATION

- . BASIC OPERATIONS (AND, OR, XOR, SHIFT)
- . CHECK EVEN/ODD
- . SWAP WITHOUT TEMP
- . SINGLE NUMBER (XOR PROPERTY)
- . SUBSETS WITH BITS

- . HAMMING WEIGHT / SET BITS
-

27. TOPOLOGICAL SORT

- . KAHN'S ALGORITHM (BFS)
 - . DFS METHOD
-

28. DISJOINT SET UNION (DSU)

- . UNION BY RANK
 - . PATH COMPRESSION
 - . CYCLE DETECTION
-

29. MINIMUM SPANNING TREE (MST)

- . KRUSKAL
 - . PRIM
 - . BORUVKA
-

30. EULER PATH

- . EULER PATH CONDITION
- . EULER CIRCUIT CONDITION

- . HIERHOLZER'S ALGORITHM
-

31. MATH

- . GCD / LCM (EUCLID)
 - . MODULAR ARITHMETIC
 - . SIEVE OF ERATOSTHENES
 - . PRIME FACTORIZATION
 - . MODULAR INVERSE
 - . FAST EXPONENTIATION
-

32. 1D DYNAMIC PROGRAMMING

- . CLIMBING STAIRS
 - . COIN CHANGE
 - . HOUSE ROBBER
 - . LIS (LONGEST INCREASING SUBSEQUENCE)
-

33. MULTI-DIMENSIONAL DP

- . KNAPSACK (0/1, UNBOUNDED)
- . MATRIX CHAIN MULTIPLICATION

- . PALINDROMIC SUBSEQUENCES
 - . LCS / EDIT DISTANCE
 - . DP ON GRID (UNIQUE PATHS, MIN PATH SUM)
-

34. KMP ALGORITHM

- . PREFIX FUNCTION (LPS ARRAY)
 - . STRING MATCHING
-

35. PRIORITY QUEUE

- . MIN PRIORITY QUEUE
- . MAX PRIORITY QUEUE
- . CUSTOM COMPARATOR
- . HEAP-BASED IMPLEMENTATION